Soft Mutation in Colloquial Welsh: A Predictive Account

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Contents

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Scope. Prescriptive standard modern **colloquial** Welsh, with Gareth King's *Modern Welsh: A Comprehensive Grammar* (3rd ed., 2016) as the normative reference. The account is strictly **predictive**: given a word and its grammatical context, does it exhibit soft mutation (SM)? Real spoken usage is not categorical (Jones, n.d.; Knight et al. 2020) — §7 records where prescription and usage genuinely diverge. The phonological *content* of mutation (t→d, p→b, c→g, …) is taken as opaque throughout; only the *trigger conditions* are at issue. Word-internal mutation under derivational prefixes is out of scope: the predicate ranges over tokens, and prefixed forms are radical lexemes in their own right.

1. The heterogeneity thesis

Every current theoretical account, whatever its framework, starts from the same premise: soft mutation has no single cause. Most instances have a lexical or morphosyntactic trigger, while a distinct residue arises under specifically syntactic conditions (Tallerman 2006). Green (2003) states the consensus most bluntly: trigger environments split into proclitic-triggered and syntax-triggered classes, and *no single generalization covers both* — the full set is arbitrary, partly item-specific, and dialectally variable. Historically this arbitrariness is the residue of reanalysis: mutation was once predictable from pure phonology (the final segment of the preceding word), but by Middle Welsh it could be predicted only from a list of triggering environments, the overwhelming majority being individual lexical items acting on the immediately following word (Willis 2007).

The program therefore implements three subsystems plus a veto layer, each returning its own provenance:

1. **Contact triggers** (`lex:*`) — a closed class of words governing a mutation grade on the immediately following dependent.
2. **Gender marking** (`gend:*`) — feminine singular nouns after the article and *un*; modifiers agreeing with a feminine singular controller.
3. **Syntactic/positional mutation** (`synt:*`) — the XP Trigger Hypothesis plus a small set of designated positions.

One constraint spans all three — the **Trigger Constraint** (Breit 2019): every mutation is licensed by the immediately preceding string element or by a feature borne by the target itself. Nothing acts at a distance. This is the strongest cross-cutting invariant in the literature and it is an invariant of the program: the environment a word is judged in is a fixed-size record about one preceding position plus the word's own resolved features, never a whole tree.

2. Reading the diagrams

Each example is rendered from the program's tree representation. Child indices appear on the connectors; every leaf is annotated with the program's verdict and its provenance. `⟨N f sg⟩` gives category and features; `lemma=` marks a trigger-lexicon key that differs from the citation form (homographs like the two *yn*, the two *ei*); `gap:NP` is an extraction site. In the Welsh text, ° marks a soft-mutated word. Verdict annotations appear on *every* leaf, so incidental facts (the colloquial mutation of the clause-initial verb, vetoes on function words) are visible alongside the example's point.

3. Contact triggers

The majority case (Willis 2007). Each trigger is a lexical entry governing a grade: full SM; *limited* SM sparing *ll-/rh-* (the article, *un*, predicative *yn*, *mor* — King §9); AM; NM; the *mixed* grade (AM on *c/p/t*, SM elsewhere — King §10); or nothing (absence is data: *pob*, *mewn*, *rhwng*, *efo*, *tair* are stored as non-triggers, not omitted). AM and NM frames matter to an SM predicate because they *pre-empt*: a word governed by an AM trigger does not soft-mutate. Blocking (King §5d) requires no rule — a trigger acts only on the immediately following word, so an intervener simply *is* the preceding element.

i °dŷ
'to a house' — the preposition i° governs SM on its dependent (King §460)

```
PP
├─0 i ⟨Other⟩ → radical (veto:no-reflex)
└─1 NP
   └─0 tŷ ⟨N m sg⟩ → SM (lex:i)
```

dy °gath
'your cat' — 2sg possessive dy° (King §111)

```
NP
├─0 dy ⟨Other⟩ immutable → radical (veto:immutable)
└─1 cath ⟨N f sg⟩ → SM (lex:dy)
```

ei °gath e — ei chath hi
'his cat' (SM) vs 'her cat' (AM, hence radical for the SM predicate) — one orthographic word, two lemmas, two grades (King §112)

```
NP
├─0 ei ⟨Other⟩ lemma=ei.3sgm → radical (veto:no-reflex)
├─1 cath ⟨N f sg⟩ → SM (lex:ei.3sgm)
└─2 e ⟨Other⟩ → radical (veto:no-reflex)
```

> With lemma `ei.3sgf` the frame grade is AM: the same tree position yields `radical (no-license)` — 'her' never soft-mutates.

fy nghath
'my cat' — fy governs NM, so the SM predicate correctly refuses (King §110)

```
NP
├─0 fy ⟨Other⟩ → radical (veto:no-reflex)
└─1 cath ⟨N f sg⟩ → radical (no-license)
```

mor °fach — mor llawn
'so small' vs 'so full' — mor° is limited SM: it spares ll-/rh- (King §105e)

```
AP
├─0 mor ⟨Adv⟩ immutable → radical (veto:immutable)
└─1 bach ⟨Adj⟩ → SM (lex:mor)
```

> Substituting llawn (ll-) at the same position yields `radical (no-license)`: the SM-ltd grade has no reflex for ll-/rh-.

mor llawn
'so full' — the ll- target resists the limited-SM trigger

```
AP
├─0 mor ⟨Adv⟩ immutable → radical (veto:immutable)
└─1 llawn ⟨Adj⟩ → radical (no-license)
```

dwy °gath — tair cath
'two cats' vs 'three cats' — dwy° governs SM, tair governs nothing (King §162)

```
NP
├─0 dwy ⟨Num⟩ → radical (no-license)
└─1 cath ⟨N f sg⟩ → SM (lex:dwy)
```

tair cath
'three cats' — feminine tair is always followed by the radical (King §162c)

```
NP
├─0 tair ⟨Num⟩ → radical (no-license)
└─1 cath ⟨N f sg⟩ → radical (no-license)
```

chwe cheffyl
'six horses' — chwe governs AM, never SM (King §162e)

```
NP
├─0 chwe ⟨Num⟩ → radical (veto:no-reflex)
└─1 ceffyl ⟨N m sg⟩ → radical (no-license)
```

chwe mlynedd — dwy °flynedd
'six years' (NM via the year-word frame) vs 'two years' (plain SM) — one lemma, two frames (King §176)

```
NP
├─0 chwe ⟨Num⟩ → radical (veto:no-reflex)
└─1 blynedd ⟨N f sg⟩ → radical (no-license)
```

> chwe carries two frames: AM generally, NM restricted to blwydd/blynedd/diwrnod. Neither yields SM. dwy °flynedd (example above) shows the same noun soft-mutating after a genuine SM trigger.

hen °ddyn
'an old man' — prenominal adjectives govern SM (King §96)

```
NP
├─0 hen ⟨Adj⟩ → radical (veto:no-reflex)
└─1 dyn ⟨N m sg⟩ → SM (lex:hen)
```

pob dyn
'every man' — pob is the one prenominal adjective that governs nothing (King §97)

```
NP
├─0 pob ⟨Other⟩ → radical (no-license)
└─1 dyn ⟨N m sg⟩ → radical (no-license)
```

cath neu °gi
'a cat or a dog' — neu° governs SM on the second conjunct (King §512)

```
NP
├─0 NP
│  └─0 cath ⟨N f sg⟩ → radical (no-license)
├─1 neu ⟨Other⟩ → radical (veto:no-reflex)
└─2 NP
   └─0 ci ⟨N m sg⟩ → SM (lex:neu)
```

y °gath neu'r ci
'the cat or the dog' — the article 'r intervenes and blocking follows from string adjacency (King §5d)

```
NP
├─0 NP
│  ├─0 y ⟨Other⟩ → radical (veto:no-reflex)
│  └─1 cath ⟨N f sg⟩ → SM (gend:art-fem-sg)
├─1 neu ⟨Other⟩ → radical (veto:no-reflex)
└─2 NP
   ├─0 y ⟨Other⟩ → radical (veto:no-reflex)
   └─1 ci ⟨N m sg⟩ → radical (no-license)
```

No blocking rule exists in the program: ’r simply IS the preceding element, and its own frame (feminine singular nouns) does not apply to masculine ci.

4. The gender subsystem

Welsh marks gender by mutation rather than affixation. Two environments: the definite article (and *un*) mutates feminine singular nouns, sparing *ll-/rh-* (King §28); and modifiers of a feminine singular controller mutate — with *no ll-/rh-* sparing (King §100: *profedigaeth °lem*), so the carve-out is category-sensitive, not phonological (Green 2003). Agreement is implemented as a feature borne by the target (Breit 2019), not as adjacency to the noun: that is what carries adjective chains, and what lets a *prenominal* adjective agree with a head that follows it. Possessor phrases are systematically immune despite string adjacency (Mittendorf & Sadler 2006; Dowle 2024) — in the program this is the *absence of a licensing relation*, derived from the genitive configuration, not a veto.

y °gath — y llong — y plant
‘the cat / the ship / the children’ — the article mutates feminine singulars only, sparing ll-/rh- (King §28)

```
NP
├─0 y ⟨Other⟩ → radical (veto:no-reflex)
└─1 cath ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

y llong
‘the ship’ — feminine singular in ll- resists the article (King §28 note b)

```
NP
├─0 y ⟨Other⟩ → radical (veto:no-reflex)
└─1 llong ⟨N f sg⟩ → radical (no-license)
```

y cathod
‘the cats’ — feminine plurals pattern with masculines: no mutation (King §28)

```
NP
├─0 y ⟨Other⟩ → radical (veto:no-reflex)
└─1 cathod ⟨N f pl⟩ → radical (no-license)
```

y °ddau °gi
‘the two dogs, both dogs’ — the article’s second frame mutates the numerals dau/dwy, which then trigger in turn (King §29)

```
NP
├─0 y ⟨Other⟩ → radical (veto:no-reflex)
├─1 dau ⟨Num⟩ → SM (lex:y)
└─2 ci ⟨N m sg⟩ → SM (lex:dau)
```

cath °fach
‘a little cat’ — adjectives agree with a feminine singular controller (King §102)

```
NP
├─0 cath ⟨N f sg⟩ → radical (no-license)
└─1 AP
   └─0 bach ⟨Adj⟩ → SM (gend:agr-mod)
```

y °ferch °fach °wen
'the little white girl' — the second adjective is not adjacent to the noun, yet mutates: agreement is a feature borne by the target (Breit 2019), not a contact effect

```
NP
├─0 y ⟨Other⟩ → radical (veto:no-reflex)
├─1 merch ⟨N f sg⟩ → SM (gend:art-fem-sg)
├─2 AP
│  └─0 bach ⟨Adj⟩ → SM (gend:agr-mod)
└─3 AP
   └─0 gwyn ⟨Adj⟩ → SM (gend:agr-mod)
```

ci mawr coch
'a big red dog' — masculine chains show no mutation anywhere; this datum forces the NP-internal exclusion on the syntactic subsystem (see §5)

```
NP
├─0 ci ⟨N m sg⟩ → radical (no-license)
├─1 AP
│  └─0 mawr ⟨Adj⟩ → radical (no-license)
└─2 AP
   └─0 coch ⟨Adj⟩ → radical (no-license)
```

y °brif °ddinas
'the capital city' — the PREnominal adjective agrees with a FOLLOWING feminine head; agreement looks rightward within the NP

```
NP
├─0 y ⟨Other⟩ → radical (veto:no-reflex)
├─1 prif ⟨Adj⟩ → SM (gend:agr-mod)
└─2 dinas ⟨N f sg⟩ → SM (lex:prif)
```

cath merch
'a girl's cat' — the possessor is immune despite string adjacency to a feminine noun (Mittendorf & Sadler 2006; Dowle 2024)

```
NP
├─0 cath ⟨N f sg⟩ → radical (no-license)
└─1 NP
   └─0 merch ⟨N f sg⟩ → radical (no-license)
```

> Derived, not stipulated: the genitive configuration [NP N NP] yields relation=possessor, and possessors are simply outside every licensing relation.

canol y °dre
'the town centre' — immunity applies to the possessor boundary, not to the possessor's inside: the article still fires within it

```
NP
├─0 canol ⟨N m sg⟩ → radical (no-license)
└─1 NP
   ├─0 y ⟨Other⟩ → radical (veto:no-reflex)
   └─1 tre ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

5. Syntactic and positional mutation

The theoretically contested residue. Two families competed: Case-based accounts, on which direct object mutation (DOM) realizes accusative Case (Zwicky 1984; Roberts 1997, 2005), and the configurational **XP Trigger Hypothesis** (Harlow 1989; Borsley 1997; Tallerman 2006), on which any phrasal category triggers SM on the word immediately following its right edge. The Case account is generally judged to both overgenerate and undergenerate (Borsley 1997; Tallerman 2006): mutation after subjects hits non-accusative material (*dim*, adverbs), and objects of nonfinite verbs mutate exactly when a phrase intervenes (Green 2003) — facts the XPTH derives for free. The program implements the XPTH geometrically: a phrasal node (never a clause) whose right edge immediately precedes the target and c-commands it licenses `synt:xp-edge`. Extraction gaps count as phrases (Willis 2007 documents the transparency lineage); following Dowle (2024) we posit no other empty categories.

Two implementation findings are worth recording as results. First, raw XPTH geometry overgenerates **NP-internally**: in a masculine chain like *ci mawr coch*, AP(*mawr*) ends immediately before *coch* and c-commands it, yet *coch* is radical. The program therefore excludes candidates whose join node with the target is an NP — Tallerman's 'complement' condition operationalized, and exactly Breit's (2019) division of labor: NP-internal mutation belongs to the gender subsystem alone. Second, clause-initial finite-verb mutation (`synt:v1-*`) is **not XPTH-derivable even in principle**: nothing precedes a clause-initial verb, and the negative variant surfaces as *mixed* mutation (AM on *c/p/t*) where the XPTH only ever produces SM. It is particle-drop residue — affirmative *fe°/mi°*, interrogative *a°*, negative *ni* (mixed) were deleted, and their mutations remained (King §481 note; Green 2003 on *ni*). The grade tracks the vanished particle exactly. In formal literary Welsh the affirmative default is particle-less and radical, so this entire rule is register-specific (King §11d) — encoded as designated positions, it is a data toggle, not a theory change. Imperatives resist both v1 mutation and *neu°* (King §512), which the category system encodes directly.

Gwelodd Mair °dŷ
'Mair saw a house' — direct object mutation: the subject NP's right edge licenses, not the verb (Harlow 1989; Borsley 1997; Tallerman 2006)

```
S
├─0 gweld ⟨V⟩ → SM (synt:v1−aff)
├─1 NP
│  └─0 Mair ⟨N⟩ immutable → radical (veto:immutable)
└─2 NP
   └─0 tŷ ⟨N m sg⟩ → SM (synt:xp−edge)
```

The verb also carries colloquial v1 SM (°Welodd) — see the v1 examples below; the verdict annotation on the verb reflects this.

Roedd dyn wedi prynu beic
'A man had bought a bike' — the object of a NON-finite verb stays radical: no phrase edge precedes it

```
S
├─0 bod ⟨V⟩ → radical (veto:no−reflex)
├─1 NP
│  └─0 dyn ⟨N m sg⟩ → radical (no−license)
├─2 wedi ⟨Prt⟩ → radical (veto:no−reflex)
└─3 VNP
   ├─0 prynu ⟨Vnoun⟩ → radical (no−license)
```

```
   └─1 NP
      └─0 beic ⟨N m sg⟩ → radical (no-license)
```

Roedd dyn wedi prynu yn y °dre °feic
'A man had bought, in town, a bike' — the decisive datum for the configurational account: an intervening PP restores mutation on the nonfinite object (Green 2003)

```
S
├─0 bod ⟨V⟩ → radical (veto:no-reflex)
├─1 NP
│  └─0 dyn ⟨N m sg⟩ → radical (no-license)
├─2 wedi ⟨Prt⟩ → radical (veto:no-reflex)
└─3 VNP
   ├─0 prynu ⟨Vnoun⟩ → radical (no-license)
   ├─1 PP
   │  ├─0 yn ⟨Other⟩ lemma=yn.loc → radical (veto:no-reflex)
   │  └─1 NP
   │     ├─0 y ⟨Other⟩ → radical (veto:no-reflex)
   │     └─1 tre ⟨N f sg⟩ → SM (gend:art-fem-sg)
   └─2 NP
      └─0 beic ⟨N m sg⟩ → SM (synt:xp-edge)
```

Beic °brynodd y °ddynes
'It was a bike the woman bought' — the fronted object has nothing before it and stays radical; a Case-based account must stipulate this (Tallerman 2006 contra Roberts 1997)

```
S
├─0 NP
│  └─0 beic ⟨N m sg⟩ → radical (no-license)
├─1 prynu ⟨V⟩ → SM (synt:xp-edge)
└─2 NP
   ├─0 y ⟨Other⟩ → radical (veto:no-reflex)
   └─1 dynes ⟨N f sg⟩ → SM (gend:art-fem-sg)
```

Pwy °welodd °ddraig?
'Who saw a dragon?' — the extraction gap occupies the subject position and counts as a phrase edge; the verb's own SM (°welodd < gwelodd) comes from the fronted wh-phrase's edge

```
S
├─0 NP
│  └─0 pwy ⟨Other⟩ → radical (no-license)
├─1 gweld ⟨V⟩ → SM (synt:xp-edge)
├─2 gap:NP
└─3 NP
   └─0 draig ⟨N f sg⟩ → SM (synt:xp-edge)
```

Rhaid i Emrys °fynd
'Emrys must go' — the PP's right edge c-commands and licenses the verbal noun; King files this under 'sentence construction' (§5e: unblockable), for the XPTH it is the same rule as DOM

```
S
├─0 rhaid ⟨N⟩ → radical (no-license)
```

```
├─1 PP
│  ├─0 i ⟨Other⟩ → radical (veto:no-reflex)
│  └─1 NP
│     └─0 Emrys ⟨N⟩ immutable → radical (veto:immutable)
└─2 VNP
   └─0 mynd ⟨Vnoun⟩ → SM (synt:xp-edge)
```

°Golles i'r tocyn
'I lost the ticket' — DOM lands on the first WORD of the object; here that is the article (no SM reflex), so the noun stays radical

```
S
├─0 colli ⟨V⟩ → SM (synt:v1-aff)
├─1 NP
│  └─0 i.pron ⟨Other⟩ → radical (veto:no-reflex)
└─2 NP
   ├─0 y ⟨Other⟩ → radical (veto:no-reflex)
   └─1 tocyn ⟨N m sg⟩ → radical (no-license)
```

°Ddylset ti °ddim
'You shouldn't' — two mutations, two subsystems: mixed particle-residue on the negative verb, subject-edge SM on dim (King §§10, 11a)

```
S (neg)
├─0 dylu ⟨V⟩ → SM (synt:v1-neg-mixed)
├─1 NP
│  └─0 ti ⟨Other⟩ immutable → radical (veto:immutable)
└─2 dim ⟨Prt⟩ → SM (synt:xp-edge)
```

Pharith hi °ddim
'It won't last' — the negative v1 grade is MIXED: AM on p-, so no soft mutation; the grade tracks the dropped particle ni (King §10)

```
S (neg)
├─0 para ⟨V⟩ → radical (no-license)
├─1 NP
│  └─0 hi ⟨Other⟩ → radical (veto:no-reflex)
└─2 dim ⟨Prt⟩ → SM (synt:xp-edge)
```

os daw e
'if he comes' — the subordinator occupies clause-initial position, so the verb is not v1 and stays radical (King §502)

```
S
├─0 os ⟨Other⟩ → radical (veto:no-reflex)
├─1 dod ⟨V⟩ → radical (no-license)
└─2 NP
   └─0 e ⟨Other⟩ → radical (veto:no-reflex)
```

°Ddwy °flynedd yn ôl aeth hi adre
'Two years ago she went home' — adverbial NPs mutate at their first word (King §11b); note the geometrically identical fronted object (dom-fronted) does NOT — the difference is authored, not derived

```
S
├─0 NP (adverbial)
│  ├─0 dwy ⟨Num⟩ → SM (synt:adv-np)
│  ├─1 blynedd ⟨N f sg⟩ → SM (lex:dwy)
│  └─2 yn ôl ⟨Adv⟩ → radical (veto:no-reflex)
├─1 mynd ⟨V⟩ → radical (veto:no-reflex)
└─2 NP
   └─0 hi ⟨Other⟩ → radical (veto:no-reflex)
```

Dewch, °blant!
'Come, children!' — vocative mutation (King §11c); the imperative verb itself stays radical: imperatives resist v1 mutation

```
S
├─0 dod ⟨Vimp⟩ → radical (no-license)
└─1 NP (vocative)
   └─0 plant ⟨N m pl⟩ → SM (synt:vocative)
```

6. Vetoes

Two ways a word can be exempt no matter what licenses it: a per-lexeme immutability flag (King §12: fixed mutations like *beth*; personal names; g-initial loanwords; miscellaneous grammatical words), and the absence of any SM reflex for the word's initial (vowels, *n*-, *s*-, *ch*-, *ff*- …, King §5a). The two are distinguished in provenance because they are different theoretical claims: one is lexical listing, the other phonological vacuity.

dy gêm
'your game' — g-initial loanwords are immutable even under a live trigger (King §12e)

```
NP
├─0 dy ⟨Other⟩ immutable → radical (veto:immutable)
└─1 gêm ⟨N f sg⟩ immutable → radical (veto:immutable)
```

i Dafydd
'to Dafydd' — personal names do not mutate in the modern language (King §12d, §36)

```
PP
├─0 i ⟨Other⟩ → radical (veto:no-reflex)
└─1 NP
   └─0 Dafydd ⟨N⟩ immutable → radical (veto:immutable)
```

i ysgol
'to school' — vowel-initial words have no SM reflex; the question does not arise (King §5a)

```
PP
├─0 i ⟨Other⟩ → radical (veto:no-reflex)
└─1 NP
   └─0 ysgol ⟨N f sg⟩ → radical (veto:no-reflex)
```

7. Contested territory

A predictive account must say where prescription itself is unstable. These are flagged in the program's contested registry; test failures in these contexts are reported as *contested*, not wrong. Corpus context: overall prescriptive application in spontaneous speech runs near 74%, no speaker is categorical, gender agreement on adjectives applies barely above chance (54%), and the particle-less finite-verb SM is the most categorical trigger measured (95%) — consistent with relexicalization of the mutated verb forms (Jones, n.d.; Knight et al. 2020).

- **euphony:s-d** — d- often fails to mutate after preceding -s: nos da, wythnos diwetha (King §12e note). King treats as euphonic override of gend:agr-mod.
- **am-erosion** — AM after â/gyda/tri frequently realized as SM or radical colloquially (King p.12; corpus: gyda 2.94% AM). We encode King's prescription (AM ⇒ no SM), but SM here is attested vernacular.
- **immutable-variation** — lle, byth: immutability 'usage varies' (King §12b).
- **v1-neg-sm** — Negative inflected verbs prescriptively take mixed mutation, but plain SM is common speech (King §10: 'more a feature of the literary language'). We encode mixed.
- **placenames** — Non-Welsh placenames prescriptively immutable (i Birmingham), but i Firmingham 'not uncommon' in speech (King §12c). After yn 'in', foreign names 'even more chaotic': yng Nghamden / yn Gamden / yn Camden all heard (King §472).
- **nm-blynedd** — NM of blynedd/blwydd after chwe and wyth 'sometimes followed by the radical': chwe blynedd, wyth blynedd (King §176 note a).
- **yn-loc-sm** — NM after yn 'in' is 'precarious at best' in speech; if any mutation heard it is usually SM (yn °Fangor, yn °Geredigion), though SM of G- names is resisted (yn Gogledd Cymru) and bare radical is common (yn Bangor). Formal written language allows none of these (King §472). Fixed exception either way: yn °Gymraeg takes SM even in the literary language (blocked NM from a lost article).
- **tan-conj** — tan as a conjunction (tan iddo ffonio) 'certainly used' in some areas but 'some speakers regard this as substandard'; standard is nes (King §467).
- **cyn-finite** — cyn + inflected verb (cyn daeth e adre) 'regarded as substandard' for cyn iddo °ddod adre (King §509).
- **tra-grade** — Degree adverb tra: King marks tra° (SM, §95); the literary tradition prescribes AM (tra chyflym). We encode King's SM.
- **am-dialect-mn** — In some dialects AM extends to radical m-/n- (m > mh, n > nh), 'regarded as non-standard' (King App A p.488). Irrelevant to the SM predicate but a pipeline hazard: mh-/nh-surface forms are not always NM.

8. The full trigger lexicon

Every contact frame in the program, with its King reference. A lemma may carry several frames (*y*, *chwe*); `none` frames record deliberate non-triggers.

Lemma	Grade	Target conditions	Source
`am`	SM	—	§448
`ar`	SM	—	§449
`at`	SM	—	§450

Lemma	Grade	Target conditions	Source
`cyn.prep`	none	—	§451 (no superscript; contrast cyn- prefix and cyn.equ)
`dan`	SM	—	§452
`o dan`	SM	—	§452
`oddidan`	SM	—	§452
`dros`	SM	—	§453
`tros`	SM	—	§453 (literary variant)
`drost`	SM	—	§453 (spoken variant)
`efo`	none	—	§454 (N ‘with’)
`gan`	SM	—	§455
`gyn`	SM	—	§455 (spoken variant of gan)
`ger`	none	—	§456
`gerbron`	none	—	§456
`gyda`	AM	—	§457; erosion CONTESTED (am-erosion)
`heb`	SM	—	§458
`hyd`	SM	—	§459
`i`	SM	—	§460
`o`	SM	—	§9
`tan`	SM	—	§467
`trwy`	SM	—	§9
`drwy`	SM	—	§9
`wrth`	SM	—	§9
`â`	AM	—	§447 (‘optionally causes AM’) CONTESTED (am-erosion)
`tua`	AM	—	App A
`yn.loc`	NM	—	App A, §472
`a.rel`	SM	—	§481
`dacw`	SM	—	App A
`dyma`	SM	—	App A
`dyna`	SM	—	App A
`fe.prt`	SM	—	App A §213
`mi.prt`	SM	—	App A §213
`yma`	SM	—	App A
`yna`	SM	—	App A
`neu`	SM	cat ∈ {N, Adj, Vnoun, Num, Adv, Prt, Other}	§512; SM cancelled before an imperative (Arhoswch neu dewch) — hence V excluded
`pan`	SM	—	App A

Lemma	Grade	Target conditions	Source
`yn.pred`	SM-ltd	cat ∈ {N, Adj, Num}	App A ('not ll-, rh-'); numerals in age/time: yn °dair oed §176
`go`	SM	—	§95
`pur`	SM	—	§95
`rhy`	SM	—	§95
`tra`	SM	—	§95 (King marks tra°; literary AM tradition) CONTESTED
`reit`	SM	—	§95
`mor`	SM-ltd	—	§105(e), App A
`cyn.equ`	SM	—	§105 (cyn °ddued)
`na.than`	AM	—	§103 (na^{h}, nag before vowels)
`fatha`	AM	—	§105 (colloquial 'like')
`fawr`	SM	—	§105(d) (fawr °gallach)
`lawn`	SM	—	§105(a) (°lawn mor °grac)
`y`	SM-ltd	cat ∈ {N}; gender=f; number=sg	§28; note (b) ll-/rh- resist
`y`	SM	cat ∈ {Num}; lexemes: dau/dwy	§29 (y °ddau, y °ddwy)
`un`	SM-ltd	cat ∈ {N}; gender=f; number=sg	§162(a)
`fy`	NM	—	§110
`dy`	SM	—	§111
`ei.3sgm`	SM	—	§112
`ei.3sgf`	AM	—	§112; colloquial omission CONTESTED (am-erosion)
`ein`	none	—	§113 (h- before vowels only)
`eich`	none	—	§113
`eu`	none	—	§113 (h- before vowels only)
`'w.3sgm`	SM	—	§112 (i'w °frawd)
`'w.3sgf`	AM	—	§112
`'w.3pl`	none	—	§112
`hen`	SM	—	§96(a)
`prif`	SM	—	§96(a)
`ambell`	SM	—	§96(a), §116
`holl`	SM	—	§96(a)
`pob`	none	—	§97 (the ONLY prenominal adjective that does not trigger SM)

Lemma	Grade	Target conditions	Source
`pa`	SM	—	§96(b)
`pwy.S`	SM	—	§96(b) (S variant of pa)
`cyn-`	SM	—	§96(c) (ex-; prefix written with hyphen but treated as trigger word)
`dirprwy`	SM	—	§96(c)
`uwch`	SM	—	§96(c)
`is-`	SM	—	§96(c)
`cryn`	SM	—	§96(d)
`unig`	SM	—	§99 (prenominal sense 'only'; postnominal 'lonely' does not trigger)
`rhyw`	SM	—	§115
`unrhyw`	SM	—	§115
`amryw`	SM	—	§116
`cyfryw`	SM	—	§116 (y cyfryw °beth)
`fath`	SM	—	§116 (y fath °beth)
`ffasiwn`	SM	—	§116
`rhai`	none	—	§115 (contrast rhyw°)
`sawl`	none	—	§187
`peth`	none	—	§193 (directly precedes sg noun, no o°, no mutation)
`dau`	SM	—	§162(b)
`dwy`	SM	—	§162(b)
`tri`	AM	—	§162(c) 'erratically in the spoken language' CONTESTED (am-erosion)
`tair`	none	—	§162(c) (always radical)
`pedwar`	none	—	§162
`pedair`	none	—	§162
`chwe`	AM	—	§162(e) 'erratically in speech' CONTESTED (am-erosion)
`chwe`	NM	lexemes: blwydd/blynedd/di-wrnod	§176; radical sometimes (chwe blynedd) CONTESTED (nm-blynedd)
`pum`	NM	lexemes: blwydd/blynedd/di-wrnod	§176
`saith`	NM	lexemes: blwydd/blynedd/di-wrnod	§176

Lemma	Grade	Target conditions	Source
wyth	NM	lexemes: blwydd/blynedd/diwrnod	§176; radical sometimes (wyth blynedd) CONTESTED (nm-blynedd)
naw	NM	lexemes: blwydd/blynedd/diwrnod	§176
deng	NM	lexemes: blwydd/blynedd/diwrnod	§176
deuddeng	NM	lexemes: blwydd/blynedd/diwrnod	§176 (12)
pymtheng	NM	lexemes: blwydd/blynedd/diwrnod	§176 (15)
deunaw	NM	lexemes: blwydd/blynedd/diwrnod	§176 (18)
ugain	NM	lexemes: blwydd/blynedd/diwrnod	§176 (20)
can	NM	lexemes: blwydd/blynedd/diwrnod	§176 (100: can °mlwydd oed)
ail	SM	—	§170(b) (both genders)
trydedd	SM	gender=f	§170(b)
pedwaredd	SM	gender=f	§170(b)
trydydd	none	—	§170(b) (masc: y Trydydd Byd)
pedwerydd	none	—	§170(b)
pumed	SM	gender=f	§170(b) (y °bumed °orsaf; masc radical)
chweched	SM	gender=f	§170(b)
seithfed	SM	gender=f	§170(b)
wythfed	SM	gender=f	§170(b)
nawfed	SM	gender=f	§170(b)
degfed	SM	gender=f	§170(b)
ugeinfed	SM	gender=f	§170(d) (yr ugeinfed °ganrif)
nos	SM	lexemes: Llun/Mawrth/Mercher/Iau/Gwener/Sadwrn/Sul/Calan	§180 (Nos °Fawrth, Nos °Galan)
mewn	none	—	§461 (non-specific 'in a'; contrast yn.loc)

Lemma	Grade	Target conditions	Source
`mo`	SM	—	§462, §295 (contraction of dim o°, carries o's SM)
`oddiar`	SM	—	§463
`oddiwrth`	SM	—	§464
`rhag`	none	—	§465 (rhag dod, rhag ofn — radical)
`rhwng`	none	—	§466 ('one of the few simple spatial prepositions which does not cause SM')
`trw`	SM	—	§468 (variant of trwy)
`wth`	SM	—	§470 (spoken variant of wrth)
`er`	none	—	§506 (er ei holl gyfoeth — no mutation from er itself)
`ers`	none	—	§503
`erbyn`	none	—	§503
`nes`	none	—	§503 (i-construction; no direct mutation)
`ar bwys`	none	—	§475 (compound preps cause no mutation on following noun)
`ar draws`	none	—	§475
`ar gyfer`	none	—	§475
`ar ôl`	none	—	§475
`er mwyn`	none	—	§475
`o flaen`	none	—	§475
`o gwmpas`	none	—	§475
`wrth ymyl`	none	—	§475
`ymhlith`	none	—	§475-476
`ymysg`	none	—	§475-476
`yn lle`	none	—	§475
`hanner can`	NM	lexemes: blwydd/blynedd/di-wrnod	§176 (50)
`a.conj`	AM	—	§510; 'often disregarded in the spoken language, particularly for th-' CONTESTED (am-erosion)
`na.nor`	AM	—	§513 (nor; 'often disregarded in speech') CONTESTED (am-erosion)

Lemma	Grade	Target conditions	Source
`na.rel`	mixed	—	§481 (NEG relative; King's exx show SM on b/d/g-initials, consistent with mixed; AM on c/p/t per prescriptive tradition + §10)
`sy`	SM	lexemes: dim	§479 (sy °ddim)
`os`	none	—	§502 (os daw e)
`pe`	none	—	§502
`hyd.conj`	none	—	§502 (hyd gwelwch chi — verb radical; contrast hyd° preposition)
`ta`	none	—	§512 (te 'ta coffi)
`ond`	none	—	§511
`ni`	mixed	—	§10
`na.neg`	mixed	—	§10
`oni`	mixed	—	§10

9. Known limits

- **Adverbial and vocative status are author-supplied.** Geometry provably underdetermines them: sentence-initial adverbial NPs mutate while geometrically identical fronted objects do not. The distinction is adjunct-vs-argument, a valence/lexical fact — every formal account consumes it as input (Dowle 2024 takes it from f-structure), none derives it.
- **Common-noun apposition** shares the genitive configuration and would be misclassified as a possessor; appositive NPs are overwhelmingly personal names (immutable), so the exposure is minimal.
- **Coordination** is a known soft spot for the XPTH generally; Welsh coordinators happen to begin with mutation-immune segments (*a*, *neu*, *ond*, *na*), which masks most of the exposure.
- **Usage variability** (§7) is out of predictive scope by design; a usage model would attach per-trigger application rates to the same rule inventory.
- The mixed-mutation grade of the negative relative particle *na* is encoded from prescriptive tradition; King's own examples (§481) are grade-ambiguous (all on *b/d/g*-initials, where mixed and SM coincide).

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